Fiscal Year:	FY 2013	**Task Last Updated:**	FY 02/28/2013
PI Name:	Wiese, Claudia Ph.D.		
Project Title:	HZE-induced skin carcinoma: an in-vivo-like multicellular epithelial model system for analysis of radiation quality effects and DSB repair processes		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	For information purposes only--PI moved in June 2014; however, funding did not go with her. As of 6/1/14, Colorado State University, Department of Environmental and Radiological Health Sciences, 485 MRB - 1618 Campus Delivery, Fort Collins, CO 80523-1618, Office: (970) 491 7618, Email: Claudia.Wiese@colostate.edu		
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
Start Date:	04/01/2011	**End Date:**	09/30/2014
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 9/30/2014 per PI (Ed., 1/28/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Snijders, Antoine (Lawrence Berkeley National Laboratory)		
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Task Description:	There is considerable evidence that ionizing radiation causes non-melanoma skin cancer (NMSC), and the risk for NMSC in astronauts is significantly increased. To further assess the cancer risk for humans from space radiation, an in-vivo-like multicellular skin model system is used. Multiple different endpoints as surrogate markers for skin cancer are assessed. Persistent chromosomal damage, altered cell proliferation and evasion from apoptosis are studied in 2D monolayer cultures of normal primary and of initiated human keratinocytes exposed to a variety of charged nuclei (Aim 1). In addition, to more realistically recapitulate the complex architecture of a tissue, an in-vivo-like skin model system, composed of fibroblasts and keratinocytes, is used to ascertain the risk of space radiation-induced cancer development processes in 3D (i.e. persistent chromosomal damage, altered cell proliferation, evasion from apoptosis, and differentiation) (Aim 2). Last, a surface transplantation assay of exposed 3D reconstructed human epithelia onto immunodeficient mice will be carried out in future to test for intermediate and late events in cancer progression (cell invasion and metastasis, respectively) (Aim 3). We hypothesize that, in basal cells, unfaithful DNA DSB repair will lead to persistent chromosomal damage, promoting altered cell growth and escape from tissue homeostasis. We believe that the proposed approach is highly relevant for assessing the risk of skin cancer in astronauts and for modeling space radiation-induced tumor development in other epithelial tissues.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Skin as a model system to study radiation-induced human epithelial cancers: Skin is one of the most complex organs of the body, consisting of multiple cell types, including epithelial keratinocytes and connective tissue (i.e. fibroblasts). Keratinocytes form a stratified squamous epithelium and are composed of one basal cell layer and several differentiated cell layers. Proliferation is limited to the basal layer, whereas differentiating cells withdraw from the cell cycle and express various differentiation markers, dependent on the stage of their differentiation. Because radiation-induced neoplasms in skin derive primarily from epithelial cells, radiobiological studies on keratinocyte monolayers (2D), as suggested by us, provide a good first approximation for assessing cancer development processes triggered by ionizing radiation, but also by other environmental mutagens. Although 2D in vitro assays are high throughput and can be very informative, they do not appropriately recapitulate the complex interactions between different cell types and the extracellular matrix within tissues. The in vivo animal model systems, may provide the complex native environment, but are not well suited for experimental manipulation and may not accurately recapitulate the human condition. Three-dimensional (3D) in vitro organotypic culture models of epithelial tissues, like the one developed by us, provide an intermediate solution by recapitulating the major features of normal 3D tissue organization using human derived cells, and hold great potential for studying the effects of radiation and other environmental mutagens on tissue homeostasis. In this model system, keratinocytes are grown on a collagen layer containing fibroblasts and are exposed to air. These conditions allow fibroblasts and keratinocytes to interact such that a stratified and differentiated epithelium is formed, resembling the architecture of the human skin. For the experiments proposed in our study, we use low-passage normal primary human keratinocytes (NPKs). The advantage of primary keratinocytes is that their response most closely resembles that of normal skin. To mimic intraepithelial neoplastic lesions, we mix NPKs with eGFP fluorescently labeled, immortalized non-tumorigenic HaCaT cells or tumor-derived SCC13 cells. HaCaT cells were derived from normal skin of a male patient and spontaneously immortalized in culture. However, HaCaT cells retain the functional differentiation properties of normal keratinocytes, form a stratified squamous epithelium when grown in organotypic cultures (despite the absence of fully functional p53), and are non-tumorigenic when injected in mice. Notably, HaCaT cells carry a UVB fingerprint mutation in p53 that resides in many skin cells of healthy middle-aged individuals. SCC13 cells were derived from a female, diagnosed with an early stage (T2N0) small facial skin tumor. SCC13 cells also carry a UVB fingerprint mutation in p53 and are characterized by complete dependence on fibroblast feeder support for culture in vitro, and by a non-invasive phenotype in vivo. Thus, while SCC13 cells were derived from a bona fide skin squamous cell carcinoma, the cells cultured from this tumor do not show signs of advanced malignancy. This model for intraepithelial neoplasia is highly relevant since it has been shown previously that adult skin contains clonal patches of p53 mutated keratinocytes approximately 60-3000 cells in size, which can involve up to 4% of the epidermis. We believe that our studies are relevant for a better understanding of the processes involved in the development of epithelial cancers due to exposure to environmental mutagens. The skin model system consisting of co-cultured fibroblasts and keratinocytes is one of the best-understood human tissue systems, and fibroblast-derived soluble factors play an important role in maintaining the well-controlled balance between epithelial proliferation and differentiation. For human skin, many of these paracrine factors and the pathways they are involved in are known, and they also are conserved across other epithelial human organs. Therefore, the response of skin tissue to environmental factors is relevant for predicting cancer risk in other tissues of epithelial origin.
Task Progress:	A human in-vivo-like multicellular skin model system is used to elucidate the mechanisms that contribute to increased risk for skin cancer from space radiation. Multiple different endpoints as surrogate markers for cancer are investigated. Persistent chromosomal damage (i.e. micronucleus (MN) formation) and altered cell proliferation are determined in 2D monolayer cultures of primary and initiated keratinocytes exposed to a variety of charged nuclei covering the linear energy transfer range from 44 to 250 keV/µm. In addition, to more realistically recapitulate the complex architecture of a tissue, 3D reconstructed skin epithelia, composed of dermal fibroblasts and keratinocytes, are used to ascertain the risk of space radiation-induced cancer development processes in 3D. We have established dose response curves of high LET Fe and Si ions for MN formation in primary and initiated 2D keratinocytes. 137Cs gamma-rays were used as low LET reference radiation. Although the dose response curves differ in shape and magnitude, exposure to 300 and 1000 MeV/n Fe ions led to the highest fraction of MN induced in both primary and initiated keratinocytes. As expected, MN formation was more efficient after 300 MeV/n Si ions than after 1000 MeV/n Si ions. Compared to primary keratinocytes, MN formation is higher in immortalized keratinocytes, reflecting the greater tolerance of persistent DNA damage in initiated cells. In 3D skin epithelia, we have assessed the effects of heavy ion exposure on cell proliferation (Ki67) and the DNA damage response (53BP1 and gamma H2AX) by immunofluorescence. In the experiments to date, there is evidence for an LET effect on cell proliferation, with a more pronounced decrease in Ki67-positive cells after exposure to 300 MeV/n Fe ions than after 1000 MeV/n Fe ions. Conversely, increased levels of unresolved gamma H2AX foci are observed in organotypic skin epithelia exposed to higher LET radiation types. We hypothesize that, in basal cells, unfaithful DNA repair in response to charged particle irradiation leads to persistent chromosomal damage, resulting in altered cells with a potentially proliferative advantage.

Bibliography Type:	Description: (Last Updated: 04/11/2018)
Abstracts for Journals and Proceedings	Snijders AM, Mannion BJ, Rydberg B, Leung SG, Kronenberg A, Wiese C. "An In-Vivo-Like Multi-Cellular Skin Model System for the Analysis of Radiation Quality Effects." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012